

Can Expert Opinion Improve Effort Predictions when Exploiting Cross-Company Datasets? - A Case Study in a Small/Medium Company

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Abstract—Many studies have shown that the accuracy of the predictions obtained by estimation models built considering data collected by other companies (cross-company models) can be significantly worse than those of estimation models built employing a dataset collected by the single company (within-company models). This is due to the different characteristics among cross-company and within-company datasets. In this paper, we propose a simple approach based on the opinion of the experts that could help in the context of small/medium company that do not have data available from past developed projects. In particular, experts are in charge of selecting data from public cross-company datasets looking at the information about employed software development process and software technologies. The proposed strategy is based on the use of a Delphi approach to reach consensus among experts involved in the selection and the use of an estimation technique to build the estimation model using the selected data. To assess the strategy, we performed an empirical study considering a dataset from the PROMISE repository that includes information on the functional size expressed in terms of COSMIC and other information on the development process to be exploited for building the cross-company estimation model. We selected this dataset since COSMIC is the method used to size the applications by the company that provided the within-company dataset employed as test set to assess the accuracy of the built cross-company model. As for the estimation technique we used simple linear regression. We compared the accuracy of the obtained predictions with those of the cross-company model built without selecting the observations and some baselines widely used in the literature. The results are promising since the obtained predictions are better than those of the other approaches.

Index Terms—Effort estimation, Cross-company estimation models, Expert opinion, Delphi approach

I. INTRODUCTION

Effort estimation is a key management activity which goes on throughout a software project being fundamental for accurate project (re)planning and for allocating resources adequately [1]. Thus, the competitiveness of a software company heavily depends on the ability of its project managers to accurately predict in advance the effort required to develop software system. Several techniques, which rely on a more formal approach, have been proposed to support project managers in estimating software development effort. These include the application of some algorithms to a number of factors that

influence the development cost, such as the size, to produce an estimate or a prediction model providing the estimation in an objective way.

The paper focuses on the problem of having an effort prediction model in the context of small/medium software companies that do not have data to build their own models [2]. In this scenario companies can approach the problem by employing estimation models obtained by considering data collected by other companies (cross-company models) and obtained from publicly available dataset (e.g., PROMISE [3]). However, many studies have shown that the accuracy of the obtained predictions can be significantly worse than those of estimation models built employing datasets collected by the company from past projects (within-company models) [4] [5] [6] [7] [8] [9] [10]. This is due to the different characteristics of the software developed by the single company or due to specific development context. Some approaches have suggested to prune data from publicly available datasets, however, they require datasets with many observations and the availability of some data of the single company as well as sophisticated learning strategies [11] [12].

In this paper, we propose a simple approach based on the opinion of the experts that could help in the context of small/medium company that do not have data available from past developed projects. In particular, company experts are in charge of selecting data looking at the information available in the public dataset about employed software development process and software technologies (e.g., programming languages). The proposed strategy is based on the use of a Delphi approach to reach consensus among experts involved in the selection and the use of an estimation technique to build the estimation model using the selected data.

To assess the strategy, we performed an empirical study considering a dataset from the PROMISE repository that includes information on the functional size expressed in terms of COSMIC and other information on the development process to be exploited for building the cross-company estimation model. This dataset was obtained by extracting information from ISBSG repository [13]. We selected this dataset since

COSMIC is the method used to size the applications by the company that provided us the test set we used to assess the accuracy of the built cross-company estimation models. As for the estimation technique we used simple linear regression (SLR) since the company involved in our study was interested in estimating the effort by exploiting only the functional size information as independent variable. Furthermore, SLR can be considered one of the most widely used estimation technique.

We indicate the strategy based on expert opinions as CCexpert, while the approach based on the use of all information contained in a publicly available dataset is named CCoriginal in the following.

To assess the accuracy of CCexpert, we compared the effort predictions obtained with it with those achieved exploiting the cross-company model built without selecting the observations and with the predictions obtained using baselines widely used in the literature.

To this aim, we defined the following research questions:

RQ₁: Are the predictions obtained with CCexpert better than those achieved with baselines?

RQ₂: Are the predictions obtained with CCexpert better than those achieved with CCoriginal?

RQ₁ was defined to verify that CCexpert is able to provide better estimations than baselines strategies like ones based on considering just the mean and the median of efforts of previously developed projects. This can be considered a sort of sanity check. Indeed, it is widely known that if an estimation technique does not provide better results than these naive models, their adoption by industry is impossible: why practitioners should spend their energy dealing with algorithmic estimation techniques if simply using the mean or the median effort of their own past projects as an estimate provides them with good enough results [31].

RQ₂ allows us to verify whether the selection performed by experts can help to improve the accuracy of the cross-company models built without carrying out selection of data from cross-company datasets.

Structure of the paper. Background on the use of cross-company effort estimation models is presented in Section II. In Section III we provide a description of the design of the performed study, while the results are presented and discussed in Section IV. Section V concludes the paper with final remarks.

II. BACKGROUND

In the last 20 years several studies have been performed to assess the usefulness of cross-company effort estimation models [14] [15] [16] [17] [18], many of them in the context of Web effort estimation which are discussed in [19] [11]. Furthermore, some literature reviews have also highlighted main results of those studies and discussed differences, e.g., [20] [10] [21]. Many of the studies found that cross-company estimation models provided results significantly worse than those of within-company estimation models. The reasons highlighted by the performed studies are mainly related to the differences between the data in the cross-company dataset

and the data in the within-company dataset. Indeed, software project data of a company can differ significantly from data of different companies, due to differences in the adopted processes and practices [4] [6]. Thus, researchers highlighted that cross-company datasets should be analyzed and partitioned in order to select the observations that could match within-company development practices and ensure a random sample representative of a well-defined population [7].

To address the above issues some recent studies proposed strategies to select the cross-company dataset more similar to the within-company dataset. Some of these proposals exploited widely used machine learning approaches, such as analogy-based technique [22] and Nearest Neighbour filtering (NN-filtering) with step-wise regression [8] [23] to select observations to be used for the cross-company effort estimation models. The results of the performed studies revealed that for the majority of the cross-company models built predictions achieved were comparable with those obtained with within-company models. However, in some cases predictions worse than those achieved with the simple baseline based on the median of previous project efforts were obtained

Other two studies [12] and [11] analyzed the use of a framework to build cross-company models to be exploited by a single company. The framework is based on a learning task that aims to highlight relationship between the single company and the companies whose data forms the cross-company dataset. In particular, the selection of the cross-company models is done by exploiting a limited number of within-company training examples. The first study validated the proposed framework exploiting data from traditional software projects [12], while the second study investigated the framework using data from Web software projects [11].

Differently, in this paper, we propose a simple approach based on the opinion of the experts that could help in the context of small/medium company that do not have data available from past developed projects. In particular, company experts are in charge of selecting data looking at the information available in the public datasets about employed software development process and software technologies (e.g., programming languages).

In the literature several studies have shown that effort predictions based only on the opinion of experts usually fail (see e.g., the review reported in [24]). However, there are some studies that have proposed to exploit opinions of experts in combination with model-based effort prediction strategies to improve. In particular, Mendes *et al.* proposed the use of Bayesian Networks (BN) to construct an expert-based Web effort model [25]. The performed study showed that the involved Web Company can benefit since the built model allowing the representation of uncertainty (which is inherent in effort estimation) outperformed expert-based estimates. Furthermore, other previous studies investigated Web effort estimation models that explicitly consider uncertainty, which is inherent to effort estimation, applying Bayesian Networks [26] [27] [28]. The analyses were based on data about 150 projects from the Tukutuku dataset to build Hybrid Bayesian Network

(BN) models (structure expert-driven and probabilities data driven) [25]. The results were encouraging since the Hybrid BN models provided results better than baselines, such as mean- and median-based effort, and widely used estimation techniques, such as multivariate regression, classification and regression trees. These analyses were successively extended by considering further data from the Tukutuku dataset and both Hybrid and data-driven different BN models [29] [30].

In our approach we involved experts in the selection of the data, about employed software development process and software technologies (e.g., programming languages), to be included in the cross-company dataset and not in the selection of the effort estimation techniques/models.

III. STUDY DESIGN

The *goal* of this study is to investigate whether the use of expert opinions for selecting data from a cross-company dataset to be employed for building cross-company estimation models can improve the accuracy of the obtained predictions.

In the following we describe the employed datasets, data selection strategy, estimation technique, validation method, and evaluation criteria and discuss threats to validity of the study.

A. Datasets

In this section we first describe the dataset from PROMISE repository used to build the cross-company effort estimation models (i.e., the training set). Then, we present the within-company dataset used to validate the obtained cross-company models and the considered baselines (i.e., the test set).

1) *Training set*: The dataset employed to build the cross-company models was selected from the PROMISE repository. For each observation it includes information on the functional size expressed in terms of COSMIC and other information on the development process that are exploited for building the cross-company estimation model. It was obtained by extracting information from ISBSG repository [13]. We selected this dataset since COSMIC is the method used to size the applications by the company that provided us the test set we used to assess the accuracy of the built cross company models.

Table I shows some summary statistics related to the applications included in the cross-company dataset. The variables are EFF, i.e., the actual effort expressed in terms of person-hours, and CFP, expressed in terms of number of COSMIC Function Points. Other information about this dataset are provided in Section III-B where we describe the selection of the cross-company dataset.

2) *Test set*: Data to validate the built cross-company models was provided by an Italian small/medium-sized software company, whose core business is the development of enterprise information systems, mainly for local and central government. Among its clients, there are health organizations, research centers, industries, and other public institutions. The company is specialized in the design, development, and management of solutions for Web portals, enterprise intranet/extranet applications (such as Content Management Systems, e-commerce,

TABLE I
DESCRIPTIVE STATISTICS OF TRAINING SET

Var	Obs	Min	Max	Mean	Median	Std. Dev.
EFF	42	40	47493	5671.6	2937	9552.55
CFP	42	2	2003	340.05	104.5	508.23

TABLE II
DESCRIPTIVE STATISTICS OF TEST SET

Var	Obs	Min	Max	Mean	Median	Std. Dev.
EFF	25	782	4537	2577	2686	988.14
CFP	25	163	1090	602.04	611	268.47

work-flow managers, etc.), and Geographical Information Systems. It is certified ISO 9001:2000, and it is also a certified partner of Microsoft, Oracle, and ESRI.

The dataset includes information on 25 applications, such as e-government, e-banking, Web portals, and Intranet applications. All the projects were developed with SUN J2EE or Microsoft .NET technologies. Oracle has been the most commonly adopted DBMS, but also SQL Server, and MySQL were employed in some of these projects. As for the collection of the information, the software company used timesheets to keep track of the development effort. In particular, each team member annotated the information about his/her development effort on each project every day, and weekly each project manager stored the sum of the efforts for the team. To collect all the significant information to calculate the values of the size measure in terms of COSMIC, a template is filled in by the project managers and the functional size is determined by the company measurers.

The summary statistics related to the 25 applications employed in our study¹ are shown in Table II.

B. Data selection strategy

To select observations, i.e., projects, from the cross-company dataset that have characteristics more close to the ones developed by the single company involved in our study, we exploited the Delphi method, which allowed us to bring together the opinions of the company experts. The Delphi method is a structured communication technique, originally developed as a systematic, interactive forecasting method which relies on a panel of experts. The experts answer questions in two or more rounds. After each round, an anonymous summary of all judgments is provided by a facilitator. Then, experts are encouraged to revise their previous answers taking into account the opinions of the other participants involved. The process can be stopped according to a predefined stop criterion (e.g., achievement of consensus, stability of results). Delphi is based on the principle that forecasts (or decisions) from a structured group of individuals are more accurate than those from unstructured groups [32]

¹Raw data cannot be revealed because of a Non Disclosure Agreement with the software company.

C. Estimation technique

To build the cross-company prediction models we have employed as an estimation technique the Simple Linear Regression (SLR), which is a model-based approach widely and successfully employed in the industrial context and in several researches to estimate development effort (see e.g., [33] [34] [35] [31] [10]).

SLR allows us to build estimation models to explain the relationship between the independent variable, denoting the employed size measure (i.e., CFP), and the dependent variable, representing the development effort (EFF). Thus, SLR allows us to obtain models of this type:

$$EFF = a + b \times CFP \quad (1)$$

where b is the coefficient that represents the amount the variable EFF changes when the variable CFP changes 1 unit, and a is the intercept. Once such a model is obtained, given a new software project for which an effort estimation is required, the project manager has to size it using the same unit of measure of the model, and to use this value in the regression equation to get the effort prediction.

Regarding the baselines we have considered in our study, we used two constant models, named C1 and C2. C1 is the constant model that exploits the mean of previous project efforts (namely, $EFF_{o_1}, \dots, EFF_{o_n}$) as the predicted effort, for each observation o_i in the dataset (for $i = 1, \dots, n$, where n is the number of observations in the cross-company dataset), while C2 takes into account the median of previous project efforts.

D. Validation method

To validate the obtained cross-company effort estimation models we have exploited two different datasets. The estimation models have been built by exploiting a first dataset (from PROMISE repository) that contains information from projects developed by different software companies in the past. A second dataset that contains information from software projects developed by a company in our research network has been used to validate the built estimation models.

E. Evaluation criteria

We decided to employ Mean of the absolute residuals (MAR) obtained for the 25 observations in the test set as evaluation criteria [36]. Furthermore, to answer our first (second, respectively) research question we applied the Wilcoxon test [37] on the distribution of the 25 absolute residuals obtained with CCexpert and the distributions of the 25 absolute residuals achieved with C1 and C2 (CCoriginal, respectively). For all the statistical tests performed in our analysis, we decided to accept a probability of 5% of committing a Type-I-Error, as is customary in Software Engineering empirical studies [38].

To have also an indication of the practical/managerial significance of the results, we verified the effect size. Effect size is a simple way of quantifying the standardized difference between two groups [39]. In particular, we employed the

TABLE III
EFFECT SIZE CLASSIFICATION

Effect size	g	Reference
Very small	0.01	[42]
Small	0.20	[40]
Medium	0.50	[40]
Large	0.80	[40]
Very large	1.20	[42]
Huge	2.0	[42]

Hedges g formula which is a somewhat more accurate version of Cohen's d since it adds a correction factor for small sample [40] [41]. Magnitudes of effect size, as initially suggested in [40] and expanded in [42], can be classified as in Table III.

F. Threats to Validity

The results of our empirical study could be affected by some threats, that should be taken into account in future investigations [43] [44].

The measurement and the collection of the information included in our datasets could affect the construct validity. As reported above, we employed a publicly available dataset included in the PROMISE repository [3]. It has been extracted from ISBSG repository, thus it respects all the criteria established by the ISBSG proponents. As an example, the selected observations are all scored as A and B regarding the data quality variable [13]. As for the dataset used as test set, it was obtained from small/medium software company in our network. The collection of the effort is done using timesheets: each team member annotated the information about his/her development effort on each project every day, and weekly each project manager stored the sum of the efforts for the team. The significant information to calculate the value of the size measure in terms of COSMIC is obtained from questionnaire filled-in by the project managers. The company measurers determine the functional size of the projects on the base of the information collected. One of the authors analyzed the filled templates and the analysis and design documents, in order to cross-check the provided information. The same author calculated the values of the size measure to cross-check the work done by the company measurers.

Concerning the conclusion validity, we carefully applied the statistical tests performed by verifying all the required assumptions.

As for the threats regarding reliability validity, all the information exploited in our study to select the observations in the training set (used to build the effort estimation model) are available on the web, i.e., PROMISE repository.

IV. RESULTS AND DISCUSSION

We first present the results about the data selection strategy introduced in Section III-B, then we describe the construction of the cross-company effort prediction models and discuss the accuracy of the obtained predictions.

A. Cross-company dataset selection

As for the selection of the cross-company dataset, three managers of the selected company were involved in the application of the Delphi method described in Section III-B. They were selected by the company taking into account their experience in managing software development process and their knowledge about software measures/metrics to summarize software process/products.

We applied the Delphi method two times. The first time we asked experts to decide what variables of the selected PROMISE dataset should be considered to select observations for the construction of the cross-company estimation model. As suggested by the method, we applied two rounds. In the first round they were asked to select which variables among those specified in the cross-company dataset have to be considered for the selection of the observations. Then, in the second round one of the authors acted as facilitator and summarized the selected variables and asked again to experts to express their opinions. Table IV shows the variables selected by the experts after the first round among those available in the cross-company dataset². At the end of the second round they decided to consider four variables: 1) Primary Programming Language (PPL); 2) Language1; 3) DBS1; and 4) Architecture.

The second time, using the same procedure of two rounds they selected the observations shown in Table V looking at the variables selected above³. These observations were selected since the values of the selected variables are close to the ones that characterize the projects that usually the company develop. Indeed, the company employs SUN J2EE or Microsoft .NET technologies for the software development. Furthermore, Oracle has been the most commonly adopted DBMS. However, also SQL Server and MySQL are employed.

From the summary of the facilitator, we can observe that many observations were not considered since they do not included information about Architecture and DBS1 (e.g., such as observations: 3199, 27927, 27913, 3133, 24302, 27917, 31448, 31240, 31451, 30879, 30719, 27547, 27537, 29893, 31512, 28129, 30611). Similarly, observations not specifying the value for Language1 were not considered. Even if observation 28020 was characterized by Java as PPL, Client server as Architecture, and Oracle as DBS1 it was not included due to the value of Language1. Observation 29834 was not selected since the values of PPL and Language1 are COBOL and Owner script, respectively. Similarly for observations 30333 and 32064 since the values for those variables are COBOL and VB.NET and COBOL and ASP.NET C#, respectively. Observation 27934 has no value for DBS1, while Language1 is equal to RPG3. So, it was not considered. Other observations were not selected due to the value of Architecture (e.g., 25271), Language1 and DBS1 (e.g., 27553), or they miss the value of DBS1 (e.g., 28884).

²see the PROMISE repository web site for the description of all the variables: <https://terapromise.csc.ncsu.edu/repo/effort/isbsg/isbsg10/isbsg-attribute-info.txt>

³see the PROMISE repository web site to see the remaining observations: <http://openscience.us/repo/effort/isbsg/cosmic.html>

It is important to mention that this second application of Delphi method to select the observations could be automated in the future, for example exploiting similarity measures and machine learning approaches. This could help to reduce the effort of the experts involved in the process and to avoid having strong dependency on expert intuition.

B. Model construction and validation

To build the cross-company effort prediction model using the data presented in Table I we first verified the linear regression assumptions, i.e., the existence of a linear relationship between the independent variable and the dependent variable (linearity), the constant variance of the error terms for all the values of the independent variable (homoscedasticity), the normal distribution of the error terms (normality), and the statistical independence of the errors, in particular, no correlation between consecutive errors (independence). To this aim, we performed statistical analyses, of course considering 95% confidence level. Both Pearson's correlation test (statistic=0.182 with p-value >0.05) [45] and the Spearman' rho test (statistic=0.234 with p-value >0.05) [37] revealed that the EFF (i.e., dependent variable) and CFP (i.e., independent variable) are not so correlated. The homoscedasticity assumption was verified by performing the Breush-Pagan Test [46]. The Shapiro test [47] revealed that the two distributions of EFF and CFP as well as the distribution of residuals (i.e., error terms) are not normal since a p-value less than 0.05 was obtained (i.e., the normality of the distribution was the null hypothesis). The Durbin-Watson test revealed that the residuals for consecutive errors are not so uncorrelated since the obtained statistic was less than 2. Taking into account the results of the performed analysis to verify linear regression assumptions, we decided to apply a log transformation to the variables [4].

The equation of the regression model obtained considering Log(CFP) as independent variable and Log(EFF) as dependent variable is (named CCoriginal):

$$\text{Log}(EFF) = 6.604 + 0.228 * \text{Log}(CFP) \quad (2)$$

and when it is transformed back to the original raw data scale we obtain:

$$EFF = 738 * CFP^{0.228} \quad (3)$$

Similarly, we obtained the regression model for the cross-company dataset resulted from the data selection strategy described in Section III-B (named CCexpert):

$$EFF = 161 * CFP^{0.456} \quad (4)$$

As for the applications of the baselines C1 and C2, the resulted predicted efforts are 5671.6 (the mean of effort) and 2937 (the median of effort), respectively.

To evaluate the prediction accuracy of CCoriginal, CCexpert, C1, and C2 and answer to research questions RQ1 and RQ2, we performed the validation as described in Section III-D employing the single-company dataset summarized in Table II. The results of the validation in terms of MAR values are reported in Table VI, while Figure 1 shows the boxplots

TABLE IV
FIRST APPLICATION OF THE DELPHI METHOD: VARIABLES SELECTED BY EXPERTS AFTER THE FIRST ROUND

Var	Description
IS	A derived field which attempts to summarize Organization Type of the project
PPL	The primary language used for the development: JAVA, C++, PL/1, Natural, Cobol etc.
Language1	The primary technology programming language used to build or enhance the software (i.e., that used for most of the build effort)
OS1	The primary technology operating system used to build or enhance the software (i.e. that used for most of the build effort)
Architecture	A derived attribute for the project to indicate if the application is: Stand alone, Multi-tier, Client server, Multi-tier with web public interface
DBS1	The primary technology database used to build or enhance the software (i.e. that used for most of the build effort), otherwise (if known) it is whether the project used a DBMS.

TABLE V
SECOND APPLICATION OF THE DELPHI METHOD: OBSERVATIONS SELECTED BY EXPERTS

obs	CFP	EFF	PPL	Language1	OS1	Architecture	DBS1
29331	210	204	Java	C++ Java	UNIX Windows NT	Multi-tier	Oracle
25081	237	600	C++	Java	Windows	Multi-tier with web public interface	MySQL
29471	61	669	Java	Java	Linux	Multi-tier with web public interface	MySQL
30658	5	784	ABAP	Java	Windows	Multi-tier with web public interface	MySQL
31895	82	903	C#	Java	Windows XP	Client server	SQL Server
27505	118	1488	Script Language	Java	Linux	Multi-tier with web public interface	MySQL
29311	643	4224	Java	Java	Windows 2000	Client server	Oracle
29310	441	47493	ASP.Net	Java 2EE	Windows XP	Client server	Oracle

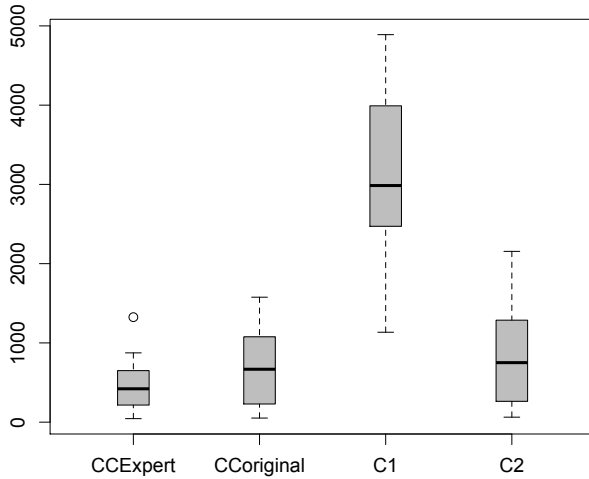


Fig. 1. Absolute residuals

of absolute residuals achieved with the four effort prediction models.

We can observe that the MAR value achieved with CCExpert is more than six times lower than the one obtained with C1 and almost two times lower than the one obtained with C2, thus highlighting significant better results for CCExpert. This results is corroborated by the tests on the statistical significance of the results (by using absolute residuals) [36]. In particular, the Wilcoxon test revealed that the estimations obtained with CCExpert are significantly better than those obtained with C1 and C2 (p-value < 0.001 in both cases) with a huge (d=3.53) and large (d=0.80) effect size, respectively.

Thus, we can positively answer to the first research question:

TABLE VI
RESULTS IN TERMS OF MEAN OF ABSOLUTE RESIDUALS (MAR)

Model	MAR
CCoriginal	701
C1	3095
C2	851
CCexpert	467

The predictions obtained with CCExpert are significant better than those achieved with two considered baselines

From Table VI we can observe that the MAR value achieved with CCExpert is more than 1.5 times lower than the one obtained with COriginal, thus highlighting much better results for CCExpert. The result of the performed Wilcoxon test revealed that the estimations obtained with CCExpert are significantly better than those obtained with COriginal (p-value < 0.001) with a medium effect size (d=0.55).

Thus, we can positively answer to the first research question:

The predictions obtained with CCExpert are significant better than those achieved with COriginal

From the results presented above we can conclude that the proposed strategy based on the opinion of experts for selecting cross-company dataset has provided interesting results for the software company involved in our study.

iiii SUGGESTIONS TO COMMUNITY llllll

V. CONCLUSION

We have proposed a simple approach based on the opinion of the experts (provided through the Delphi method) to build cross-company effort prediction models. In particular, experts are in charge of selecting data from public cross-company datasets looking at the information about employed software development process and software technologies, before to apply an estimation technique to build the prediction models. This strategy could help in the context of small/medium company that do not have data available from past developed projects. The performed empirical study, involving a software company that exploits COSMIC as functional size measurement method, has provided encouraging results. Indeed, the effort predictions obtained with the proposed strategy are significantly better than those obtained with the model built on the whole cross-company dataset.

As future work, we have planned to replicate this study with other publicly datasets and with other companies of our research network. We also intend to further assess the proposed strategy considering different estimation techniques, e.g., when software companies are interested to build prediction models based on several independent variables. Moreover, as mentioned above, we could automate the selection of the observations to be included in the cross-company dataset for the construction of the estimation model, after the selection of the variables through the Delphi approach. To this aim, similarity measures and machine learning approaches could be exploited.

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